

Gender, Race, and Diet Affect Platelet Function Tests in Normal Subjects Contributing to a High Rate of Abnormal Results

Summary To assess sources of variability in platelet function tests in normal subjects, 64 healthy young adults were tested on 2–6 occasions at 2 week intervals using 4 methods: platelet aggregation (AGG) in platelet-rich plasma (PRP) in the Bio/Data PAP-4 Aggregometer (BD) and Chrono-Log Lumi-Aggregometer (CL); and AGG in whole blood (WB) in the CL and Multiplate Platelet Function Analyzer (MP), with ATP release (REL) in CL-PRP and CL-WB. Food and medication exposures were recorded prospectively for 2 weeks prior to each blood draw. At least one AGG abnormality was seen in 21% of 81 drug-free specimens with CL-PRP, 15% with CL-WB, 13% with BD-PRP, and 6% with MP-WB, increasing with inclusion of REL to 28% for CL-PRP and 30% for CL-WB. Epinephrine AGG and REL were significantly reduced in males ($P<0.0001$). Ristocetin AGG and collagen and thrombin REL were significantly reduced in Blacks ($P<0.0001$). One-third of specimens drawn following flavonoid-rich food exposures had aberrant results, compared to 8.5% of specimens without such exposures ($P=0.0035$). PRP tests had less intra-individual variation than WB tests. Gender, race, diet, and test system affected results of platelet function testing in healthy subjects, suggesting caution when interpreting the results of platelet function testing in patients.